Sharded, Isolated, Sliced, and Aggregated (SISA) Exact Machine Unlearning (Bourtoule et al. 2021)

Exact removal of a training point's influence from a deep neural network without full retraining requires architectural partitioning (SISA), trading ensemble accuracy and storage overhead for bounded retraining cost.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

In standard stochastic gradient descent, every training example updates shared parameters across all subsequent batches, making exact post-hoc removal of a single data point equivalent to retraining from scratch unless the training pipeline partitions data into isolated shards and checkpointed slices (SISA).

EVIDENCE AND CAUSAL ASSESSMENT

Bourtoule et al. (2021) prove that partitioning a dataset into S disjoint shards and R sequential slices per shard bounds the retraining work of a single deletion request to one shard from the slice where the point first appeared, yielding exact distributional equivalence to a model trained without that point.

EFFECT SIZE AND PERSISTENCE

On Purchase, SVHN, and ImageNet benchmarks, SISA reduces retraining time per unlearning request by $2.45\timesto $4.63× over naive retraining, but degrades ensemble accuracy when shard count S becomes large or when deletion requests arrive non-uniformly across shards.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as robust (<20%) for exact unlearning under bounded request rates. However, adaptive or adversarial deletion streams that hit every shard collapse SISA's speedup back toward full retraining.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Approximate gradient-ascent or representation-editing "unlearning" heuristics on monolithic LLMs do not provide the certified deletion guarantee of SISA or (\varepsilon, \delta)-differential privacy; suppressed facts can often be recovered via rephrasing or few-shot fine-tuning.

WHAT ACTION IS REASONABLY JUSTIFIED

Regulated AI systems subject to statutory deletion mandates (GDPR Art. 17) should separate immutable base representations from partitioned, revocable retrieval indexes or sharded adapters rather than relying on unverified weight-editing heuristics.

VERIFIED SOURCES & DOI IDENTIFIERS

- Bourtoule, L., Chandrasekaran, V., Choquette-Choo, C. A., Jia, H., Travers, A., Zhang, B., Lie, D., & Papernot, N. (2021). Machine Unlearning. 2021 IEEE Symposium on Security and Privacy (SP), 141–159. https://doi.org/10.1109/sp40001.2021.00019

- Dwork, C., McSherry, F., Nissim, K., & Smith, A. (2006). Calibrating Noise to Sensitivity in Private Data Analysis. Theory of Cryptography (TCC 2006), LNCS 3876, 265–284. https://doi.org/10.1007/11681878_14